KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Sampling Calculations

Name _____

SA1

Complete each stratified sampling table.

Complete the stratified sample sizes.A researcher wants a stratified sample of 78 from a population of 780.

Group	Population	Sample
Year 11	180	?
Year 7	240	?
Year 8	210	?
Year 9	150	?

Show your working

KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Sampling Calculations

Name _____

SB1

Find the interval and first selected positions for each systematic sample.

Find the sampling interval and list the first five selected positions.A list contains 720 names. A systematic sample of 36 names is required. The random starting position is 17.

Show your working

KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Sampling Calculations

Name _____

SC1

Describe a valid simple random sampling process for each situation.

Describe a valid simple random sampling process.A researcher needs to choose 50 people from a list of 960 people.

Show your working

KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Sampling Calculations

Name _____

SD1

Choose a suitable sampling method and justify the choice.

Choose a suitable sampling method and justify your choice.A researcher wants to investigate working-from-home arrangements for all employees in a company.

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Sampling Calculations

Name _____

EA1

Complete each stratified sampling table.

Complete the stratified sample sizes.A researcher wants a stratified sample of 120 from a population of 960.

Group	Population	Sample
Year 7	320	?
Year 10	280	?
Year 8	240	?
Year 11	120	?

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Sampling Calculations

Name _____

EB1

Find the interval and first selected positions for each systematic sample.

Find the sampling interval and list the first five selected positions.A list contains 500 names. A systematic sample of 20 names is required. The random starting position is 2.

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Sampling Calculations

Name _____

EC1

Describe a valid simple random sampling process for each situation.

Describe a valid simple random sampling process.A researcher needs to choose 20 people from a list of 360 people.

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Sampling Calculations

Name _____

EA1

Complete each stratified sampling table.

Complete the stratified sample sizes.A researcher wants a stratified sample of 120 from a population of 960.

Group	Population	Sample
Year 7	320	?
Year 10	280	?
Year 8	240	?
Year 11	120	?

Show your working

Teacher answers & working

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SA1

Complete each stratified sampling table.

Complete the stratified sample sizes.A researcher wants a stratified sample of 78 from a population of 780.

Group	Population	Sample
Year 11	180	?
Year 7	240	?
Year 8	210	?
Year 9	150	?

Answer:

Year 11: 18, Year 7: 24, Year 8: 21, Year 9: 15

Working:

Use the stratified sampling formula for each group.

Group sample size = $\frac{group population}{total population}$ × total sample size

Year 11: $\frac{180}{780}$ × 78 = 18

Year 7: $\frac{240}{780}$ × 78 = 24

Year 8: $\frac{210}{780}$ × 78 = 21

Year 9: $\frac{150}{780}$ × 78 = 15

The completed sample is: Year 11: 18, Year 7: 24, Year 8: 21, Year 9: 15.

SB1

Find the interval and first selected positions for each systematic sample.

Find the sampling interval and list the first five selected positions.A list contains 720 names. A systematic sample of 36 names is required. The random starting position is 17.

Answer:

Interval 20; first five positions: 17, 37, 57, 77, 97

Working:

In systematic sampling, divide the population size by the required sample size.

Sampling interval = $\frac{720}{36}$ = 20

Start at the random starting position, 17.

Selected positions: 17, 17+20, 17+2(20), 17+3(20), 17+4(20)

The first five selected positions are 17, 37, 57, 77, 97.

Teacher answers & working

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SC1

Describe a valid simple random sampling process for each situation.

Describe a valid simple random sampling process.A researcher needs to choose 50 people from a list of 960 people.

Answer:

Number the people using 3-digit labels, then use a random number generator to select 50 different labels.

Working:

Start with a complete sampling frame: the list of all people in the population.

Give each person a unique 3-digit number, such as 001 to 960.

Use a random number generator to select 50 different numbers from the list.

Ignore repeated numbers and numbers outside the valid range if they occur.

The people with the selected numbers form a simple random sample because each person had an equal chance of selection.

SD1

Choose a suitable sampling method and justify the choice.

Choose a suitable sampling method and justify your choice.A researcher wants to investigate working-from-home arrangements for all employees in a company.

Answer:

Simple Random Sampling because it gives every member of the population an equal chance, provided the sampling frame is complete.

Working:

Identify the target population and the information available.

Target population: all employees in a company.

A suitable method is simple random sampling because it gives every member of the population an equal chance, provided the sampling frame is complete.

The justification must connect the method to representativeness, fairness, or practicality.

Teacher answers & working

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EA1

Complete each stratified sampling table.

Complete the stratified sample sizes.A researcher wants a stratified sample of 120 from a population of 960.

Group	Population	Sample
Year 7	320	?
Year 10	280	?
Year 8	240	?
Year 11	120	?

Answer:

Year 7: 40, Year 10: 35, Year 8: 30, Year 11: 15

Working:

Use the stratified sampling formula for each group.

Group sample size = $\frac{group population}{total population}$ × total sample size

Year 7: $\frac{320}{960}$ × 120 = 40

Year 10: $\frac{280}{960}$ × 120 = 35

Year 8: $\frac{240}{960}$ × 120 = 30

Year 11: $\frac{120}{960}$ × 120 = 15

The completed sample is: Year 7: 40, Year 10: 35, Year 8: 30, Year 11: 15.

EB1

Find the interval and first selected positions for each systematic sample.

Find the sampling interval and list the first five selected positions.A list contains 500 names. A systematic sample of 20 names is required. The random starting position is 2.

Answer:

Interval 25; first five positions: 2, 27, 52, 77, 102

Working:

In systematic sampling, divide the population size by the required sample size.

Sampling interval = $\frac{500}{20}$ = 25

Start at the random starting position, 2.

Selected positions: 2, 2+25, 2+2(25), 2+3(25), 2+4(25)

The first five selected positions are 2, 27, 52, 77, 102.

Teacher answers & working

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EC1

Describe a valid simple random sampling process for each situation.

Describe a valid simple random sampling process.A researcher needs to choose 20 people from a list of 360 people.

Answer:

Number the people using 3-digit labels, then use a random number generator to select 20 different labels.

Working:

Start with a complete sampling frame: the list of all people in the population.

Give each person a unique 3-digit number, such as 001 to 360.

Use a random number generator to select 20 different numbers from the list.

Ignore repeated numbers and numbers outside the valid range if they occur.

The people with the selected numbers form a simple random sample because each person had an equal chance of selection.